

UR5-TRON

6-DOF Robotic Arm

Design, Mechanical Analysis & Engineering Documentation

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Independent Research Project

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1 Project Overview

This document constitutes the engineering and design record for TRON, a six-degree-of-freedom (6-DOF) serial robotic arm developed as my independent research project. The work presented herein spans the complete design cycle: from kinematic configuration and mechanical layout, through joint-level static torque analysis and actuator sizing, to a fully justified and self-consistent mechanical specification. Every numerical result is traceable to first principles, and every design decision is evaluated against a uniform Safety Factor of 1.5.

The primary objective of the project was to produce a compact, desktop-scale manipulator capable of handling a useful payload defined here as 250 to 500 g within a total reach envelope of 512 mm from the base pivot to the end-effector tip. The arm was to be realised using commercially available stepper actuators and planetary gearboxes, without recourse to custom-wound actuators or belt-and-pulley transmission systems. All joints are driven either by a geared stepper assembly or by direct-drive stepper actuators, depending on the torque and mass requirements of the specific joint.

All mass properties are parametrically modelled against the density of 3D-printed PLA. The documented kinematic lengths and inertia matrices represent the nominal V2 architecture, providing a baseline for the static torque analysis prior to physical manufacturing.

The designation TRON reflects the design philosophy of the project: a structurally transparent, mechanically honest manipulator whose operation can be understood, reproduced, and extended. The red anodised aluminium housings and carbon-tube links are not merely aesthetic choices; they reflect the material selection decisions documented in the design record.

System Specification Summary

2 Mathematical Framework

2.1 Rigid-Body Statics and the Torque Equation

The torque required at a revolute joint is determined by the sum of moments produced by all masses located distal to that joint, evaluated under the worst-case configuration. For a planar arm fully extended along the horizontal, the gravitational moment about joint i is:

$$\tau_i = g \cdot \sum_j m_j \cdot d_{ij} \quad (1)$$

where:

- τ_i : required holding torque at joint i [N·m]

Parameter	Value
Degrees of Freedom	6 (Base, Shoulder, Elbow, Wrist Pitch, Wrist Yaw, End Effector)
Total Arm Mass (estimated)	1.5 kg
Rated Payload	250 – 500 g (worst-case analysis at 500 g)
Link 1: Ground → Shoulder pivot	90 mm
Link 2: Shoulder → Elbow pivot	170 mm
Link 3: Elbow → Wrist Pitch pivot	170 mm
Link 4: Wrist Pitch → End-Effector tip	82 mm
Total reach (ground to tip)	512 mm
Safety Factor (all joints)	$SoF = 1.5$
Gearbox efficiency (planetary)	$\eta = 0.85$
Gravitational acceleration	$g = 9.81 \text{ m/s}^2$

- g : gravitational acceleration, 9.81 m/s^2
- m_j : mass of the j -th distal body [kg]
- d_{ij} : horizontal distance from joint i to the centre of mass of body j [m]

This formulation assumes quasi-static operation, meaning accelerations are sufficiently small relative to gravity that inertial terms are negligible in the dominant load case. Dynamic effects are accounted for implicitly through the Safety Factor.

2.2 Safety Factor

A Safety Factor of 1.5 is applied uniformly across all six joints. The design torque, the minimum torque that the drive system must be capable of producing, is therefore:

$$\tau_{\text{design}} = 1.5 \cdot \tau_{\text{required}} \quad (2)$$

This factor provides margin against dynamic loading during acceleration and deceleration, manufacturing and assembly tolerances, friction in structural bearings beyond the gearbox efficiency figure, and degradation over the operational life of the mechanism. A value of 1.5 is standard practice for light-duty robotic manipulators operating in controlled environments.

2.3 Actuator Output Torque

For a stepper actuator driving through a planetary gearbox, the output torque at the joint shaft is:

$$\tau_{\text{output}} = \tau_{\text{actuator}} \cdot R \cdot \eta \quad (3)$$

where:

- τ_{actuator} : rated holding torque of the stepper actuator [N·m]
- R : gear reduction ratio (dimensionless)
- η : mechanical efficiency of the gearbox (0.85 for planetary type)

For a direct-drive configuration, $R = 1$ and $\eta = 1$, giving $\tau_{\text{output}} = \tau_{\text{actuator}}$. The acceptance criterion for each joint is:

$$\tau_{\text{output}} \geq \tau_{\text{design}} = 1.5 \cdot \tau_{\text{required}} \quad (4)$$

3 Kinematic Geometry and Mass Distribution

3.1 Link Lengths and Reference Frame

The arm is modelled as a four-link open kinematic chain, with link lengths defined as the distances between consecutive joint axes. The following values were measured from the CAD model and are used as the basis of all subsequent calculations.

Link Designation	Length
L_1 : Ground to Shoulder pivot	90 mm = 0.090 m
L_2 : Shoulder to Elbow pivot	170 mm = 0.170 m
L_3 : Elbow to Wrist Pitch pivot	170 mm = 0.170 m
L_4 : Wrist Pitch to End-Effector tip	82 mm = 0.082 m
Total reach ($L_1 + L_2 + L_3 + L_4$)	512 mm = 0.512 m

3.2 Mass Distribution

The total arm mass is 1.5 kg, distributed across the structural segments as follows. Masses are estimated from component weights (actuator + gearbox + housing + link tube) and represent the midpoint of the expected range for each assembly.

Segment	Estimated Mass
Base joint (actuator + 27:1 gearbox + housing + base plate)	0.550 kg
Upper arm link (carbon tube + hardware)	0.080 kg
Shoulder joint assembly (actuator + 27:1 gearbox + bracket)	0.350 kg
Forearm link (carbon tube + hardware)	0.080 kg
Elbow joint assembly (actuator + 19:1 gearbox + bracket)	0.280 kg
Wrist pitch (NEMA 17 42mm + 3:1 gear + housing)	0.120 kg
Wrist yaw + End Effector (two Pancake actuators + structure)	0.040 kg
TOTAL	1.500 kg

For the purpose of torque calculations, the mass of each link is treated as a point load acting at the geometric midpoint of that link. Joint assembly masses are treated as point loads acting at the joint pivot location.

4 Joint Torque Analysis

The following sections present the static torque analysis for each of the six joints. In every case, the worst-case posture is assumed: the arm is fully extended in the horizontal plane, maximising the moment arms of all distal masses. This represents the configuration in which gravitational loading is greatest, and constitutes the governing load case for actuator sizing.

Numerical constants applied throughout: $g = 9.81 \text{ m/s}^2$, $SoF = 1.5$, $\eta = 0.85$ (planetary gearbox)

4.1 Joint 1: Base Rotation

The base joint rotates the entire arm about the world vertical (Z) axis. Gravity acts along this axis and therefore produces no static bending moment at the base pivot when the arm is upright. The dominant load is the inertial torque required to accelerate and decelerate the full arm assembly. To produce a conservative static-equivalent estimate, the arm is modelled as a

horizontal cantilever emanating from the base pivot, with all masses acting at their horizontal distances.

Component	Contribution [N·m]
Upper arm link CoM ($d = 0.175$ m)	$0.080 \times 0.175 = 0.0140$
Shoulder joint ($d = 0.090$ m)	$0.350 \times 0.090 = 0.0315$
Elbow joint ($d = 0.260$ m)	$0.280 \times 0.260 = 0.0728$
Forearm link CoM ($d = 0.345$ m)	$0.080 \times 0.345 = 0.0276$
Wrist pitch ($d = 0.430$ m)	$0.120 \times 0.430 = 0.0516$
Wrist yaw + EE ($d = 0.471$ m)	$0.040 \times 0.471 = 0.0188$
Payload ($d = 0.512$ m)	$0.500 \times 0.512 = 0.2560$
$\Sigma(m \cdot d)$	$0.4723 \text{ N}\cdot\text{m} / (g)$

$$\tau_1 = g \cdot \Sigma(m_j \cdot d_{1j}) = 9.81 \times 0.4723 = 4.633 \text{ N} \cdot \text{m} \quad (5)$$

Applying the Safety Factor:

$$\tau_{1_design} = 1.5 \times 4.633 = 6.950 \text{ N} \cdot \text{m} \quad (6)$$

Available output torque for NEMA 17 ($\tau_m = 0.40 \text{ N} \cdot \text{m}$) with 27:1 planetary ($\eta = 0.85$):

$$\tau_{1_output} = 0.40 \times 27 \times 0.85 = 9.180 \text{ N} \cdot \text{m} \quad (7)$$

Base joint (27:1) verified for 500 g payload. Margin: +32%

4.2 Joint 2: Shoulder

The shoulder joint elevates and positions the entire arm above it. It is the most heavily loaded joint in the manipulator, as it must support the combined gravitational moment of the upper arm, elbow, forearm, wrist, and payload at maximum horizontal extension.

Component	Contribution [N·m]
Upper arm link CoM ($d = 0.085$ m)	$0.080 \times 0.085 = 0.0068$
Elbow joint ($d = 0.170$ m)	$0.280 \times 0.170 = 0.0476$
Forearm link CoM ($d = 0.255$ m)	$0.080 \times 0.255 = 0.0204$
Wrist pitch ($d = 0.340$ m)	$0.120 \times 0.340 = 0.0408$
Wrist yaw + EE ($d = 0.381$ m)	$0.040 \times 0.381 = 0.0152$
Payload ($d = 0.422$ m)	$0.500 \times 0.422 = 0.2110$
$\Sigma(m \cdot d)$	$0.3418 \text{ N}\cdot\text{m} / (g)$

$$\tau_2 = 9.81 \times 0.3418 = 3.353 \text{ N} \cdot \text{m} \quad (8)$$

Applying the Safety Factor:

$$\tau_{2_design} = 1.5 \times 3.353 = 5.030 \text{ N} \cdot \text{m} \quad (9)$$

Available output torque for NEMA 17 with 27:1 planetary ($\eta = 0.85$):

$$\tau_{2_output} = 0.40 \times 27 \times 0.85 = 9.180 \text{ N} \cdot \text{m} \quad (10)$$

9.180 N·m $\geq$ 5.030 N·m. Shoulder (27:1) verified. Margin: +82.5%

4.3 Joint 3: Elbow (Design Comparison 19:1 vs. 27:1)

The elbow joint articulates the forearm relative to the upper arm. The distal assembly is substantially lighter than the full arm above the shoulder, and the moment arms are shorter.

Component	Contribution [N·m]
Forearm link CoM ($d = 0.085$ m)	$0.080 \times 0.085 = 0.0068$
Wrist pitch ($d = 0.170$ m)	$0.120 \times 0.170 = 0.0204$
Wrist yaw + EE ($d = 0.211$ m)	$0.040 \times 0.211 = 0.0084$
Payload ($d = 0.252$ m)	$0.500 \times 0.252 = 0.1260$
$\Sigma(m \cdot d)$	$0.1616 \text{ N}\cdot\text{m} / (g)$

$$\tau_3 = 9.81 \times 0.1616 = 1.585 \text{ N} \cdot \text{m} \quad (11)$$

Applying the Safety Factor:

$$\tau_{3_design} = 1.5 \times 1.585 = 2.377 \text{ N} \cdot \text{m} \quad (12)$$

Comparison Analysis at the Elbow: I selected the 19:1 at the Elbow, justified on mass,

Parameter	19:1 Reducer	27:1 Reducer
Output torque ($\eta = 0.85$)	$0.40 \times 19 \times 0.85 = 6.460 \text{ N}\cdot\text{m}$	$0.40 \times 27 \times 0.85 = 9.180 \text{ N}\cdot\text{m}$
τ_{design} ($SoF = 1.5$)	$2.377 \text{ N}\cdot\text{m}$	$2.377 \text{ N}\cdot\text{m}$
Torque margin	+172%	+286%
Typical mass (planetary)	~ 120 g	~ 200 g
Effect on shoulder τ	+0.200 N·m on shoulder	+0.333 N·m on shoulder
Output speed (relative)	Higher (more responsive)	Lower
Acceptance	PASS	PASS (over-specified)

speed, and cost grounds.

4.4 Joint 4: Wrist Pitch

The wrist pitch joint tilts the distal wrist assembly up and down. This joint is driven by a NEMA 17 (42 mm) stepper actuator via a 3:1 gear reducer.

Component	Contribution [N·m]
Wrist yaw + EE assembly CoM ($d = 0.041$ m)	$0.040 \times 0.041 = 0.0016$
Payload ($d = 0.082$ m)	$0.500 \times 0.082 = 0.0410$
$\Sigma(m \cdot d)$	$0.0426 \text{ N}\cdot\text{m} / (g)$

$$\tau_4 = 9.81 \times 0.0426 = 0.418 \text{ N} \cdot \text{m} \quad (13)$$

Applying the Safety Factor:

$$\tau_{4_design} = 1.5 \times 0.418 = 0.627 \text{ N} \cdot \text{m} \quad (14)$$

Available output torque NEMA 17 (42 mm) with a 3:1 gear ratio ($\eta = 0.85$):

$$\tau_{4_output} = 0.40 \times 3 \times 0.85 = 1.020 \text{ N} \cdot \text{m} \quad (15)$$

Wrist Pitch (3:1 gear reducer) verified for 500 g. Margin: +63%

4.5 Joint 5: Wrist Yaw

The governing load is the inertial resistance of the end-effector assembly and payload during angular acceleration. For a compact end-effector and payload assembly ($m_{\text{total}} = 0.540$ kg) modelled as a point mass at radius $r = 0.040$ m from the yaw axis, and assuming a representative angular acceleration $\alpha = 10$ rad/s²:

$$\tau_{5_inertial} = m \cdot r^2 \cdot \alpha = 0.540 \times (0.040)^2 \times 10 = 0.0086 \text{ N} \cdot \text{m} \quad (16)$$

Applying the Safety Factor:

$$\tau_{5_design} = 1.5 \times 0.0086 = 0.013 \text{ N} \cdot \text{m} \quad (17)$$

Available output torque Pancake NEMA 17, direct drive:

$$\tau_{5_output} = 0.10 \text{ N} \cdot \text{m} \quad (18)$$

Wrist Yaw Pancake verified. Margin: +669%

4.6 Joint 6: End Effector

For a grip force $F_{\text{grip}} = 5$ N applied through a rack-and-pinion with pinion radius $r_p = 10$ mm:

$$\tau_6 = F_{\text{grip}} \times r_p = 5 \times 0.010 = 0.050 \text{ N} \cdot \text{m} \quad (19)$$

Applying the Safety Factor:

$$\tau_{6_design} = 1.5 \times 0.050 = 0.075 \text{ N} \cdot \text{m} \quad (20)$$

Available output torque Pancake NEMA 17, direct drive:

$$\tau_{6_output} = 0.10 \text{ N} \cdot \text{m} \quad (21)$$

End Effector Pancake verified. Margin: +33%

5 Summary of Torque Analysis

Table 5.1 consolidates the torque analysis for all six joints. For each joint, the table presents the calculated static torque, the design torque after application of $SoF = 1.5$, the selected drive configuration, the available output torque, and the resulting margin.

Joint	τ_{calc} (N·m)	τ_{design} (N·m)	Drive System	τ_{output} (N·m)	Margin
Base (J1)	4.633	6.950	NEMA17 + 27:1	9.180	+32%
Shoulder (J2)	3.353	5.030	NEMA17 + 27:1	9.180	+82.5%
Elbow (J3)	1.585	2.377	NEMA17 + 19:1	6.460	+172%
Wrist Pitch (J4)	0.418	0.627	NEMA17 + 3:1	1.020	+63%
Wrist Yaw (J5)	0.0086	0.013	Pancake direct	0.100	+669%
End Effector (J6)	0.050	0.075	Pancake direct	0.100	+33%

6 Justified Design Decisions

6.1 Base: 27:1 Planetary with NEMA 17

The base joint rotates the arm in yaw about the world vertical axis. I selected the 27:1 planetary with NEMA 17 because it provides ample rotational stiffness, positioning resolution, and torque capability for this role. As shown in section 4.1, the verified safe payload at the base easily handles the maximum 500 g requirement, keeping the base joint robust and highly capable against rotational inertia.

6.2 Shoulder: 27:1 Planetary with NEMA 17

The shoulder is the governing joint of the manipulator. It supports the largest gravitational moment of any joint in the chain, and a failure at the shoulder results in an uncontrolled drop of the entire arm. The 27:1 planetary was selected to provide a substantial torque reserve. This margin is not conservative excess; it is structurally warranted. During dynamic operation, the shoulder experiences impulsive loads during acceleration that can transiently exceed the quasi-static requirement. The 27:1 absorbs these peaks without stalling.

6.3 Elbow: 19:1 Planetary with NEMA 17

Both the 19:1 and 27:1 reducers satisfy the elbow torque requirement with substantial margin. I selected the 19:1 on the basis of three independent engineering arguments. First, the torque margin of 172% is more than sufficient. Second, the 80 g mass reduction relative to the 27:1 option reduces the moment arm acting on the shoulder joint, directly lowering the shoulder torque requirement. Third, the higher output speed of the 19:1 improves elbow responsiveness, which is important for the arm's agility.

6.4 Wrist Pitch: 3:1 Gear Reducer with NEMA 17 (42 mm)

The wrist pitch joint is driven through a compact 3:1 gear reducer to provide sufficient torque for the 500 g payload while maintaining a low mass profile. This prevents adding excessive weight to the distal assembly, which would propagate torque requirements backward through the entire kinematic chain. No belt transmission is employed anywhere in this design.

6.5 Wrist Yaw and End Effector: Pancake NEMA 17 Direct Drive

I specified the Pancake NEMA 17 for the wrist yaw and end-effector joints on the basis of its minimal axial envelope and low mass. The inertial torque requirement at the wrist yaw is less than 0.013 N·m, and the end-effector grip torque requirement is 0.075 N·m, both well within the 0.10 N·m capability of the pancake actuator. The choice of direct drive at these joints is both mechanically appropriate and mass-optimal.

7 Conclusions

The torque analysis presented in this document validates my actuator configuration for TRON against a Safety Factor of 1.5 for a 500 g payload and a 512 mm reach. The results confirm that all six joints satisfy the design criterion with margins ranging from 32% to 669%.

The gear ratio decisions are individually justified on engineering grounds. The 27:1 ratio at the base and shoulder provides the structural reserve appropriate for the most critical joints. The 19:1 at the elbow reflects a considered balance between torque availability, joint mass, output speed, and cost. The 3:1 reduction and direct-drive configurations at the wrist reflect the mass sensitivity of the distal assembly and the adequate torque margin available from the chosen actuators.

The following work items constitute the next phase of the project:

1. Finite element analysis of the main structural links under the calculated load cases.
2. Derivation of the Denavit-Hartenberg parameter set for the kinematic model.
3. Selection and integration of rotary encoders for closed-loop position control.
4. Physical prototype build and empirical validation of the torque calculations.

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